

No. 2, 60CL Street, Quarter 9, Cat Lai Ward
Thu Duc City, Ho Chi Minh City, Vietnam

+49 17 673 973 907

✉ hong.nguyen@wias-berlin.de

📄 www.wias-berlin.de/~nguyen

Full name: Nguyễn Quân Bá Hồng

Ho Chi Minh City, Vietnam

nguyenquanbahong@gmail.com

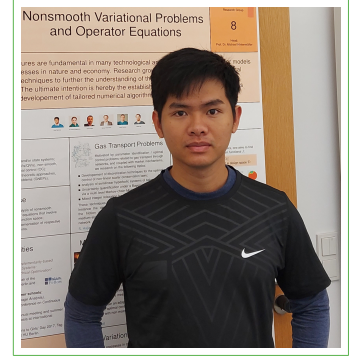
hong.nguyenquanba@umt.edu.vn

[GitHub/nqbh.github.io](https://github.com/nqbh)

[Google Site/nguyenquanbahong](https://www.google.com/search?q=Google+Site/nguyenquanbahong)

[Google Scholar/Hong Quan Ba Nguyen](https://scholar.google.com/citations?user=HONGQUANBA)

[ResearchGate/Hong Quan Ba Nguyen](https://www.researchgate.net/profile/Hong-Quan-Ba-Nguyen)



Hong Nguyen

Curriculum Vitae

"The essence of mathematics lies in its freedom." - Georg Cantor

"It is true that a mathematician who is not somewhat of a poet, will never be a perfect mathematician." - Karl Weierstrass

Education

- 2020–2021 **PhD student** enrolled in **Humboldt-Universität zu Berlin**, *Weierstrass Institute for Applied Analysis & Stochastics* (WIAS), Berlin, Germany.
In the Research Group **Nonsmooth Variational Problems & Operator Equations**, Head: Prof. **Michael Hintermüller**, within the European Innovative Training Network **Reduced Order Modelling, Simulation & Optimization of Coupled Systems** (ROMSOC), in the Project 11 **Optimal Shape Design of Air Ducts in Combustion Engines**.
- 2018–2019 **Master de Sciences, Technologie, Santé, Mathématiques et Applications**, *Université de Rennes 1*, Rennes, France, *assez bien*.
Master 2 in Fundamental Mathematics & Applications within ENS Rennes program (skipped Master 1).
- 2014–2018 **Bachelor of Science in Mathematics**, *University of Science* (HCMUS), *Viet Nam National University*, *Ho Chi Minh City* (VNU HCM), Vietnam, *very good*.
Specialized in Mathematical Analysis & Numerical Analysis.
- 2011–2014 **Elementary Mathematics**, *Ben Tre High School for Gifted Students* (CBT), Vietnam, *very good*.

Master Thesis

- Title *Staggered & well-balanced discretization of shallow-water equations*. [[pdf](#)][[slide](#)]
- Supervisor Prof. **Nicolas Seguin**
- Description This thesis investigates some staggered well-balanced finite volume schemes for the shallow water equations. A new staggered scheme, called "offset equilibrium", is developed to conquer the *dry-wet transition* issues where almost all the traditional finite volume schemes collapse. Higher-order extensions for this staggered scheme are also mentioned.

Master Courses

- 2018–2019 **Courses of Master 2 Program in Fundamental Mathematics with Applications**, 2018–2019:
- Sep 2018 Prof. **Nicoletta Tchou**. *Sobolev spaces & elliptic equation*.
- Sep 2018 Prof. **Florian Méhats**. *Hyperbolic equations*.
- Nov 2018 Prof. **Nicolas Seguin** & **Eric Darrigrand**. *Finite Element Method (FEM)*.
- Nov 2018 Prof. **Nicolas Seguin**. *Numerics of transport*.
- Nov 2018 Prof. **Mohammed Lemou**. *Numerical methods for kinetic transport equations*.
- Dec 2018 Prof. **Luis Miguel Rodrigues**. *Master 2 Seminar*
▷ Hong Nguyen. "On the smallest constant for a Gagliardo-Nirenberg functional inequality". [[pdf](#)][[slide](#)]
- Mar 2019 **Vincent Duchêne**. *Modeling & mathematical analysis of shallow free-surface fluid flows*.
- Mar 2019 Prof. **Marc Quincampoix**. *Optimal control & Hamilton-Jacobi equations*.

Research Interests

- **Mathematical Analysis** ▷ Partial Differential Equations (PDEs): Porous Medium equations (PME), Navier–Stokes equations (NSEs), Differential Geometry (DG).
- **Numerical Analysis** ▷ Finite Difference Methods (FDMs), Finite Element Methods (FEMs), Finite Volume Methods (FVMs).

- **Mathematical Optimization** ▷ Optimal Control, Shape Optimization, Topology Optimization.
- **Fluid Mechanics** ▷ Fluid Dynamics, Computational Fluid Dynamics (CFDs).
- **Turbulence** ▷ Turbulence modeling.
- **Machine Learning (ML) & Deep Learning (DL)** ▷ Computer Music, Computer Vision.

Experience

Vocational

- Jan 2025–now **Lecturer**, *University of Management & Technology (UMT)*, Ho Chi Minh City.
Department of Information Technology: Mathematics for Data Science & Artificial Intelligence.
- Jan 2022–Dec 2024 **Self-employed**, Ben Tre city, Vietnam.
Educator & Writer.
- Apr 2021–Dec 2021 **Software Development Engineer (SDE)**, *Math+Tec GmbH*, Vienna, Austria.
Secondments for technical & scientific training at the industrial partner within ROMSOC project.
- Mar–Jun 2019 **Master Internship**, *Institut de recherche mathématique de Rennes – IRMAR*, Rennes, Bretagne, France.

Miscellaneous

- 2015–2017 **Olympiad Trainer**, *Ben Tre High School for Gifted Students (CBT)*, Vietnam.
Trainer of the selected team of CBT for Vietnamese Mathematical Olympiads (VMOs).

Publications

- [math/PDE] Nguyen Anh Dao, Jesús Ildefonso Díaz, Quan Ba Hong Nguyen. “Pointwise Gradient Estimates in Multi-dimensional Slow Diffusion Equations with a Singular Quenching Term”. *Adv. Nonlinear Stud.* (2020). DOI <https://doi.org/10.1515/ans-2020-2076>. [pdf]

Preprints

Computer Science

- [CS/IS] Nguyen Quan Ba Hong. “A Functional Level-Set Slack Identity: Elevating Empirical OOD Baselines into Exact Continuous Combinatorial Equations”. [submitted]
- [CS/IS] Nguyen Quan Ba Hong. “Alexandrov Completions, Simplicial Nerves, and Ahlswede–Zhang Obstructions over Finite Covering Spaces”. [submitted]
- [CS/IS] Nguyen Quan Ba Hong. “Homological Quantification of the Čech Defect: Quillen Fibrations and Ahlswede–Zhang Obstructions over Finite Posets”. [submitted]
- [CS/IS] Nguyen Quan Ba Hong. “Boundary Optimization for Rough Sets: Continuous Integral Decoupling and Intractability Barriers”. [submitted]
- [CS/alg-cmp] Nguyen Quan Ba Hong. “A Measure-Theoretic Complexity Dichotomy for Combinatorial Up-Set Summations”. [submitted]

Mathematics

- [math/EA] Nguyen Quan Ba Hong. “Algorithmic Reduction of Elementary Symmetric Inequalities via Invariant Positivstellensatz Certificates”. [submitted]
- [math/NT] Nguyen Quan Ba Hong. “An Automated Resolution and Generalization of Rowland’s Binomial Divisibility Conjectures via Algebraic Rank Minimization”. [submitted]
- [math/PDE] Nguyen Quan Ba Hong. “Universal Gradient Estimates and Exact Free Boundary Profiles for the Singular Porous Medium Equation”. [submitted]
- [math/PDE] Nguyen Quan Ba Hong. “Spatial Decoupling and Complete Quenching for Singular Porous Medium Equations”. [submitted]
- [NA/FVM] Nguyen Quan Ba Hong. “Convergence and Asymptotic Commutativity of the Adjoint PISO Algorithm on Polyhedral Meshes”. [submitted]

Physics

Pedagogy/Education

- [mathEdu] Nguyen Quan Ba Hong. “Beyond Code Generation: The Cognitive Necessity of Analytic Derivation in Mathematical Olympiad Training in the Era of Generative AI”. [revising to resubmit]
- [mathEdu] Nguyen Quan Ba Hong. “Fundamental Asymptotic Limits of Generative AI in Discrete Mathematical Reasoning”. [revising to resubmit]

Books & Other Writings

- [bk1] Nguyen Quan Ba Hong. “*Theoretical Competitive Programming*”. [preparing]
[bk2] Nguyen Quan Ba Hong. “*Programmable Mathematical Olympiads*”. [preparing]

Courses

- Oct 12–23, 2020 Thematic Einstein Semester (TES): [Energy-based mathematical methods for reactive multiphase flows](#).
▷ TES Student Compact Course: [Variational Methods for Fluids & Solids](#).
Apr–Jun 2020 Dr. [Stephan Schmidt](#). *Computational Geometry & Shape Optimization*. Master course, Humboldt Universität zu Berlin, 2020.

Seminars

- Mar 2020–now [Berliner Oberseminar Nichtlineare partielle Differentialgleichungen \(Langenbach-Seminar\)](#), Research Group 1: *Partial Differential Equations*, WIAS Berlin, Germany.
Mar 2020–now [Material Modeling Seminar](#), WIAS Berlin, Germany.
Mar 2020–now [Numerical Mathematics Seminars](#), Research Group 3: *Numerical Mathematics & Scientific Computing*, WIAS Berlin, Germany.
Mar 2020–now [Modern Methods in Applied Stochastics & Nonparametric Statistics](#), Research Group 6: *Stochastic Algorithms & Nonparametric Statistics*, WIAS Berlin, Germany.
Mar 2020–now [Joint Research Seminar on Mathematical Optimization](#), Research Group 8: *Non-smooth Variational Problems & Operator Equations*, WIAS Berlin, Germany.
Mar 2021–now [Optimization, Control & Inverse Problems](#), Research Group 8: *Non-smooth Variational Problems & Operator Equations*, WIAS Berlin, Germany.
Mar 2020–now [Berlin Mathematics Research Center MATH⁺ Events](#).
Nov 2020–Apr 2021 ROMSOC Scientific ESRs Seminar.

Schools

- Jul 26–30, 2021 [Summer Schools on Applied Mathematics and Mechanics 2021 \(SAMM 2021\)](#): *Shape & Topology Optimization*. [GAMM Juniors](#), Graz University of Technology (Technische Universität Graz), Graz, Austria.
Jul 1–2, 2021 [Durham Days of Analysis & PDE Summer School](#). Durham University, United Kingdom.
Sep 7–10, 2020 [Model Order Reduction Summer School 2020 \(MORSS2020\)](#). École polytechnique fédérale de Lausanne (EPFL), Lausanne, Switzerland.
Jul 20–24, 2020 [162nd European Study Group with Industry](#). University of Leeds, United Kingdom.

Workshops

- Apr 29, 2021 [FUHRI2021](#): Finite volUme metHods for Real-world appLications, WIAS Berlin, Germany.
Sep 28–29, 2020 ROMSOC Internal Workshop.
Sep 15, 2020 ROMSOC Ethical Monitoring - 3rd Workshop on Ethics.

Conferences

- Mar 24–25, 2021 Annual Meeting of the Second Phase of the DFG Priority Programme 1962 ([SPP1962](#)): *Non-smooth & Complementarity-based Distributed Parameter Systems: Simulation & Hierarchical Optimization*. Deutsche Forschungsgemeinschaft (DFG) + WIAS, Germany.
Sep 14–17, 2020 [Deutsche Mathematiker-Vereinigung](#) (German Mathematical Society) Annual Meeting 2020 ([DMV2020](#)), Technische Universität Chemnitz, Chemnitz, Germany.

Talks

- Apr 12, 2021 Hồng Quân Bá Nguyễn (WIAS Berlin). “*Optimal Shape Design of Air Ducts in Combustion Engines: Design a General Shape Optimization Framework*”. ROMSOC ESR Seminar. [[abstract](#)]
Sep 7, 2020 Hong Nguyen (WIAS Berlin). “*A shape optimization problem for stationary Navier-Stokes flows in three-dimensional tubes*”. [Model Order Reduction Summer School 2020 \(MORSS2020\)](#), École polytechnique fédérale de Lausanne (EPFL), Lausanne, Switzerland. [[abstract](#)][[slide](#)]
Jun 26, 2019 Hong Nguyen. “*Staggered & well-balanced discretization of shallow water equations*”. PhD interview, Department of Mathematics, RWTH Aachen University, Aachen, Germany. [[slide](#)]

Miscellaneous

- Feb 22, 2021 [WIAS-Day2021](#) ▷ [Hong Quan Ba Nguyen \(Research Group 8\)'s Research Profile](#).
- Nov 30, 2020 [ROMSOC Blog](#) ▷ Hong Nguyen. [Optimal shape design of air ducts](#) [pdf].
- Jun 18, 2020 [Prof. Luis Miguel Rodrigues's recognition of Master equivalence & admission to doctoral studies](#) of NGUYEN Quan Ba Hong. IRMAR, Rennes, France.
- May 27, 2019 [Prof. Luis Miguel Rodrigues's recommendation letter](#) for NGUYEN Quan Ba Hong. IRMAR, Rennes, France.

Visits

- Apr 2021–Nov, 2021 [MATH+TEC GmbH](#), Vienna, Austria.
- Apr 2021–Nov 2021 [Vienna University of Technology/Technische Universität Wien](#), Vienna, Austria.
- Apr 23, 2021 [University of Vienna/Universität Wien](#), Vienna, Austria.
- Mar 2020–Nov 2021 [Humboldt-Universität zu Berlin](#), Berlin, Germany.
- Jun 25–27, 2019 [Department of Mathematics, Rheinisch-Westfälische Technische Hochschule \(RWTH\) Aachen University](#), Aachen, Germany.
- Sep 2018–Jun 2019 [UFR Mathématiques, Université de Rennes 1](#), Rennes, Bretagne, France.
- Sep 2018–Dec 2018 [Université Rennes 2 - UFR Sciences sociales](#), Rennes, Bretagne, France.
- Mar 2017 [Vietnam Institute for Advanced Studies in Mathematics \(VIASM\)](#), Hanoi, Vietnam.

Awards & Scholarships

- Mar 2020–Aug 2021 [Marie Skłodowska-Curie Actions \(MSCA\) PhD Fellowship](#), *Horizon 2020 - Innovative Training Network - European Industrial Doctorate Programs*, Grant Agreement number: 765374 - [Reduced Order Modelling, Simulation & Optimization of Coupled Systems](#) (ROMSOC) [H2020-MSCA-ITN-2017-EID/ROMSOC/ESR11].
- 2018–2019 [Lebesgue Master Scholarship](#) for *Master 2* program within ENS Rennes program.
- 2017 Scholarship of [Vietnam Institute for Advanced Studies in Mathematics](#) (VIASM).
- 2016 Special Prize ⇐ First Prize in Algebra + First Prize in Analysis, in [Vietnamese Mathematical Olympiad for College Students](#) (VMC 2016).
- 2015 Second Prize in Algebra, Second Prize in Analysis, in [Vietnamese Mathematical Olympiad for College Students](#) (VMC 2015).
- 2014 First Prize #1 in Ben Tre city Mathematical Olympiad for High School Students.
- 2013 Second Prize in Ben Tre city Mathematical Olympiad for High School Students.
- 2012 Gold Medal ranked #2, in [Vietnamese Apr 30th Mathematical Olympiad for Central & Southern Vietnam](#).

Computer Skills

- GitHub [GitHub/NQBH](#) ▷ [GitHub/ROMSOC project](#).
- Basic R, Fortran, Fireshape, Firedrake, FEniCS, Python 3, STAR-CCM+.
- Intermediate Antigravity CLI, git, OpenFOAM.
- Advanced C/C++, UNIX/Linux [openSUSE, (K)Ubuntu], T_EX/L^AT_EX, MATLAB.

Languages

- Vietnamese Mother tongue
- English Intermediate

Collaborators

In alphabetical order:

Students

In alphabetical order:

- [1] NGUYỄN NGỌC THẠCH (2026): Software Engineering ⊂ Current: Software Engineer at Zalo, VNG Corporation, Vietnam.

Referees

In chronological order:

- [1] **Dr. Đào Mạnh Khang** Bachelor Teacher
- [2] **Dr. Nguyễn Tấn Trung** Bachelor Teacher

Hobbies

- Psychology
- Literature
- Arts (emotions carriers): movies, animes, mangas, (instrumental) musics, etc.
- Philosophy
- Blogging



[Updated to Aug 11, 2026]